

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 178

THE QUINN THE PINWHEEL SUPPRESSION DISC

Slide it, flick it, it curls
CO2 pools at floor level and flows



the shuffleboard rule

ONE-HAND DISC · DELAYED RELEASE · GAS CARPET

ATTACK THE BASE OF THE FIRE

Fire system — v1.0 in force

GP-IM-178

Revision v1.0 — 23 August 2026

179 of 290

VETTED BATCH 21 — PASS · FIELD TESTS OWED

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 178

PHASE 178: THE QUINN THE PINWHEEL SUPPRESSION DISC — SLIDING CO2 FIREBREAK (THE SHUFFLEBOARD RULE) — STATUS: CURRENT — PHYSICAL-PRINCIPLE PASS; RANGE, DISCHARGE RATE AND SUPPRESSION TIME FIELD TESTS OWED. Fire suppression for Earth and gravity, delivered like shuffleboard. The existing answer fails by rolling: extinguisher balls roll past where you throw them, and any slope or stairs rules them out. Phase 178 is the disc: half a small CO2 extinguisher — about 4 inches across, 1.5 thick, one-hand manageable — on small heat-rated metal bearing wheels, with a press-button delayed release (the flashbang mechanism class) and pinwheel spin-venting. Slide it down the corridor; flick it and it curls around corners like a curling stone; and at 1.5 inches high it slides under a burning car, a forklift, a pallet stack — where no crew can crawl. Then gravity does the delivery: CO2 is one and a half times the density of air, so released at floor level it pools and flows along the ground, into the fire's base — the moving firebreak you walk behind as the flames drop. The car-rescue triage, in his blunt logic: kids inside, driver down — throw one in, get the kids out; they are breathing the top air while the disc works the floor. Aboard the fleet the confined space acts in the disc's favor — a pod floods to suppression concentration fast. The vet passed the physical principle; range, discharge rate and suppression time are the field tests, in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-178, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the disc law as seated (contents line, the bodies — the problem, the device, the slide, the gas carpet, the doctrine — the audit and provenance, verbatim) — §2 why density and slide (graded) — §3 the system on the slide — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 178: The Pinwheel Suppression Disc — Sliding CO2 Firebreak (The Shuffleboard Rule). A one-hand disc (4in x 1.5in, a quarter-kilo CO2 charge — the disc's honest volume) with pinwheel spin-venting, button-delayed release, top grab handle, and heat-rated bearing wheels, slid like a shuffleboard/curling stone — flickable around corners, low enough to reach under burning vehicles, forklifts, pallets and racking. Floor-level release exploits CO2's 1.5x air density: the gas pools, flows under obstructions, and floods down stairwells to unreachable fire bases, laying a moving firebreak corridor crews walk behind to attack or escape. Fireworks-factory table boxes (anti-static bonded variant) slow cascades; aboard ship, factory pods stage discs at Phase 163 breather-paired stations.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Unreachable-Base Fire Deaths Where Crews Cannot Approach — Delivery by Slide, Not by Throw '

Fire Suppression Baseline: Mobile Low-Profile Gas Delivery for Corridors, Vehicles & Confined Pods

Live instruction, 30 July 2026. Fifth invention of the sitting — fire suppression for Earth and gravity environments, with confined-pod application aboard. Draws on Archer's pyrotechnics trade (the pinwheel reference is his). Physics audit appended per file doctrine.

THE PROBLEM — FIRE BALLS ARE THEIR OWN ENEMY (VERBATIM)

- **The existing answer fails by rolling:** current extinguisher balls keep rolling past where you throw them — any slope or stairs rules them out entirely, and a fire's base is so often exactly where a ball will not stay: under the burning car, inside the pallet stack, at the foot of the stairwell.
- **The approach problem:** flames come up a stairwell and no crew ever reaches the base; forklift and equipment fires radiate heat no one walks into; warehouse racking shields its own fire. Suppression that cannot be delivered is suppression that does not exist.

THE DEVICE (VERBATIM)

- The disc: half the size of a small CO2 extinguisher — about 4 inches across, 1.5 inches thick, as large as is manageable with one hand. A larger cousin of the disc-shaped flash and smoke-release grenades, built on the fireworks pinwheel principle: angled vents that spin the body as gas releases, laying a radial carpet instead of a point burst. The disc's internal volume (~310 cubic centimetres at 4in x 1.5in) holds about a quarter kilo of liquid CO2 — roughly an eighth of a small 2 kg extinguisher's charge.
- **The release:** press-button delayed release — the same proven mechanism class as the flashbangs. Set the delay, slide, and the disc begins venting as it travels away from the operator.

- **The running gear:** a grab handle on top, and small built-in round bearing wheels (metal, heat-rated — no plastic near a fire) so the disc slides flat and true like a shuffleboard stone.

THE SLIDE — SHUFFLEBOARD, CURLING, AND CORNERS (VERBATIM)

- **Delivery is a skill anyone already has:** slide it down the corridor like a shuffleboard bowl — or curling, where sweepers run in front; same concept, minus the sweepers. The disc's flat profile and bearing wheels make the path predictable — unlike a ball, it goes where it is sent and stops where friction says.
- **Flick it and it curls:** impart spin on release and the disc curves — around a corridor corner, into a room, off a wall. The same physics as the curling stone's curl; the operator's wrist is the guidance system.
- **The low profile is the weapon:** 1.5 inches slides under a burning car where not even a fire crew can get, under a forklift, under pallets and racking in a factory or warehouse — straight to the base of the fire, where the fuel is.

THE GAS CARPET — GRAVITY DOES THE DELIVERY (VERBATIM)

- **CO2 is one and a half times the density of air.** Released at floor level it pools and flows along the ground — under the vehicle, into the pallet stack, and downhill. On a stairwell the carpet flows DOWN the stairs to the base of the fire that no crew member can approach. The disc releases at the one altitude where the gas itself becomes the delivery system.
- **The moving firebreak:** in a corridor of high heat, slide the disc ahead and start walking as the CO2 hits and the flames drop — then attack with an extinguisher through the window it bought, or run through. It does not promise extinguishment in open air; it promises a survivable corridor for the seconds that matter, and that promise is enough to save lives on.
- **The car-rescue triage, in Archer's blunt logic:** car on fire, kids inside, driver dead or unconscious — throw one in, get the kids out quickly. They are breathing smoke already; added CO2 in a seconds-long extraction is moot against the smoke that is killing them now. Rescue doctrine agrees: the present poison outranks the added one.

DEPLOYMENT DOCTRINE (VERBATIM)

- **Fireworks factories:** smaller units with the same release mechanism, a dozen in a box on every table. You may not stop the cascade — but you may slow it enough to save lives. In a flash-fire trade, seconds are the whole game. (Audit flag: anti-static bonding is mandatory here — CO2 discharge can generate static, and the fireworks bench is the one place a spark cannot be risked; the factory variant is bonded and grounded by design.)

- **Warehouses, factories, stairwells, vehicles:** staged at racking aisles, stair tops, and loading docks — anywhere the base of a fire hides from a hose.
- **Aboard the fleet:** the confined space acts in the disc's favor — a pod floods to suppression concentration fast. Sleep pods and crew rooms carry little flammable load; factory pods will, and factory pods get discs at the Phase 163 stations. The fleet's own law applies without exception: CO2 in a closed environment is also an asphyxiant — the disc lives on the same station as the breathers, per the pairing Archer set when the medical pods were built.

PHYSICS NOTE — THE AUDIT (KIMI: AFFIRMED — DENSITY, SLIDE, AND THE HONEST WINDOW) (VERBATIM)

Three physical claims anchor the device and all three hold. First, density: CO2 at roughly 1.5 times the density of air genuinely pools at floor level and flows downhill — low-altitude release converts gravity into a distribution pump, and suppression chemistry (flames fail below roughly 15% oxygen for most fuels) does the rest where the gas collects. Second, the slide: a flat disc on low-friction bearings travels predictably where a sphere wanders — rolling resistance on a line contact, no off-axis roll degree of freedom — and spin-induced curve (the curling stone's lateral friction asymmetry) is real, measurable, and already used by every curler on Earth; the pinwheel venting adds gyroscopic steadiness to the same path. Third, the honest window: in open air the carpet dilutes in seconds to tens of seconds — the disc is a knockdown-and-pass device, not an extinguishment guarantee, and Archer's own framing says exactly that ('attack with extinguisher or run through', 'slow it enough to save lives'). The claim is calibrated; the audit affirms it. Flags per doctrine: discharge chills surfaces toward dry-ice temperatures (handles and any skin-contact zones are insulated); in occupied enclosures the CO2-breathing conflict from the fleet's Phase 163/165 doctrine applies in full — station pairing with breathers is mandatory, and Earth fire codes already treat total-flood CO2 in occupied space as lethal-dose territory for the same reason; and the fireworks-factory variant carries anti-static bonding because a suppression device that can spark is no suppression device. The ball critique is documented fact: spherical throw-activated extinguishers are on the market, they roll where physics takes them, and slopes defeat them — the disc exists because the base of a fire is so often a place a ball refuses to stay.

ORIGIN OF THOUGHT — PHASE 178 (VERBATIM)

Archer, 30 July 2026: "fires suppression earth based and gravity as well, half size of a small extinguisher co2, if you have ever seen disc shaped flash grenades or gas/smoke release discs this is a larger spinning device, back to the fireworks pinwheel, press button for delayed release, grab the handle on the top and slide it down the corridor like a shuffleboard bowl like they play where someone or 2 are the sweepers in front as it slides, you get the concept, high heat in a corridor, it could have small built in round bearing wheels, slide start walking as co2 hits and flames drop, attack with extinguisher or run through. flick

around a corner or into a room, roll over to a burning forklift or piece of equipment roll it under a burning car where not even a fire crew can get to. under pallets and racking in a factory or warehouse."

Archer, 30 July 2026: "smaller units with the same release mech as the flashbangs should be a dozen in a box on every table in a fireworks factory, you may not stop the cascade but you may slow it enough to save lives. also good for stairwell fires because flames come up and you are never getting to the base of the fire, the existing balls versions are their own enemy it will keep rolling not stop where you throw it, any slope or stairs is out. car on fire kids inside driver dead or unconscious, throw one in, get kids out quickly, they are breathing smoke now CO₂ is moot. might be handy to throw in a pod where the confined space act in its favour, whilst sleep pods/crew rooms will have little flammable components, factory pods will, 4 niches across, 1.5 inches thick as large as is manageable with one hand."

Why it matters, in plain English: every fire death from an unreachable base is a delivery failure, not a suppression failure — the chemistry existed, the path did not. A ball rolls away from the job; a disc slides to it, ducks under it, and lets the heaviest honest gas in the box crawl into the places no firefighter can go. It buys seconds on purpose and says so — and seconds, in fire, are lives.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE DENSITY PHYSICS, AFFIRMED: CO₂ at 1.5x air density genuinely pools and flows at floor level — a gas carpet delivered to the fire's base, which is exactly where flames feed and exactly where crews cannot reach. Gravity does the delivery; the disc is just the courier.

THE SLIDE, AFFIRMED: a low flat disc on bearing wheels goes where a ball cannot — under vehicles, under racking, around corners with spin. Shuffleboard and curling are skill-sets every hand already has; the delivery training is pre-installed.

THE DELAYED RELEASE, AFFIRMED AS THE RIGHT MECHANISM CLASS: set the delay, slide, and it vents as it travels — the suppression arrives with the disc and continues after it stops. Proven mechanism, new duty.

THE HONEST WINDOW (the audit's bound): a disc is a firebreak and an access tool, not a guaranteed kill — deep-seated or shielded fires need the follow-up attack, and the fire's own doctrine says so: slide, walk in behind the carpet, then finish with the extinguisher. The vet's field-test rider lives in §9.

§3 — THE SYSTEM ON THE SLIDE

- THE DISC: ~4in x 1.5in, a quarter-kilo CO₂ charge — the disc's honest volume, one-hand manageable; a larger cousin for the big jobs.

- THE WHEELS: small built-in round bearing wheels, metal, heat-rated — no plastic near a fire.
- THE RELEASE: press-button delayed — the flashbang mechanism class; vents as it travels.
- THE VENTING: pinwheel spin pattern — the gas carpet spreads as the disc moves.
- THE GRAB: top handle for the pick-up and the slide.
- THE SMALLS: fireworks-factory units, a dozen in a box on every table — slow the cascade enough.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE BASE-OF-THE-FIRE DOCTRINE: suppression belongs at the base; the disc is the only delivery that slides there.
- THE WALK-BEHIND DOCTRINE: the firebreak moves and the crew advances behind it — attack follows the carpet.
- THE TRIAGE DOCTRINE: breathable top air, suppressed floor — the rescue window is bought, not found.
- THE CONFINED-SPACE DOCTRINE: aboard the fleet, the pod's small volume is the disc's ally — concentration arrives fast.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the disc: body, wheels, valve, the delayed release — mechanism class provenance documented.
- STEP 2 — Bench the venting: discharge rate, spin pattern, carpet spread and depth on the flat.
- STEP 3 — Field the slide (the vet's yellow): range, curl control, under-vehicle and around-corner delivery.
- STEP 4 — Field the suppression: pan fires, pallet fires, vehicle undersides — suppression time logged.
- STEP 5 — Publish the disc, the delay spec, and the deployment cards — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 187 — the backdraft bayonet (GP-IM-187): the through-door companion — the disc slides the corridor, the bayonet takes the door.

- PHASE 152 — the blast-dampening valves (GP-IM-152): pressure-event kin aboard the fleet.
- PHASE 163 — the casualty pods (GP-IM-163): the rescue triage doctrine's shipboard half.
- PHASE 173 — the fire law (GP-IM-173): elimination first; the disc is for the fires that happen anyway.

§8 — DO-NOT-BUILD

- DO NOT roll it — balls roll past; the disc slides and stops where the fire is.
- DO NOT plastic the running gear — metal, heat-rated; the delivery zone is the fire.
- DO NOT trust it as the kill — it is the firebreak and the access; the attack follows behind.
- DO NOT vent to face height — the carpet works the floor; keep the gas low.

§9 — OWED

OWED (the vet's yellow): range, discharge rate, and suppression-time field testing — slide delivery envelope, carpet spread rates, pan/pallet/vehicle fire results (§6).

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-178, v1.0 — CURRENT — PHYSICAL-PRINCIPLE PASS; range, discharge rate and suppression-time field tests owed, seated law, master v2.1 (numerical print order). Built 23 August 2026, eleventh of the 20-chapter 'ok another 20 lets keep this train rollin' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587 and 590): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 178